Status	Finished
Started	Thursday, 15 October 2026, 7:30 PM
Completed	Thursday, 15 October 2026, 7:35 PM
Duration	4 mins 44 secs
Grade	**15.00** out of 15.00 (**100**%)

Question **1**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm is known for its simplicity and sequential comparison of elements?

a. Heap Sort
b. Quick Sort
c. Bubble Sort
d. Merge Sort

The correct answer is: Bubble Sort

Question **2**

Complete

Mark 1.00 out of 1.00

Which of the following is not an in-place sorting algorithm?

a. Heap sort
b. Quick sort
c. Selection sort
d. Merge sort

The correct answer is:
Merge sort

Question **3**

Complete

Mark 1.00 out of 1.00

Which Python function would you use to sort a list in-place?

a. arrange()
b. sorted()
c. sort()
d. order()

The correct answer is: sort()

Question **4**

Complete

Mark 1.00 out of 1.00

What is mean by stable sorting algorithm?

a. A sorting algorithm is stable if it preserves the order of duplicate keys
b. A sorting algorithm is stable if it preserves the order of non-duplicate keys
c. A sorting algorithm is stable if it doesn't preserver the order of duplicate keys
d. A sorting algorithm is stable if it preserves the order of all keys

The correct answer is:
A sorting algorithm is stable if it preserves the order of duplicate keys

Question **5**

Complete

Mark 1.00 out of 1.00

Which of the following best describes the process of Merge Sort?

a. It repeatedly finds the minimum element and moves it to the sorted part of the list
b. It compares adjacent elements and swaps them if necessary
c. It builds a sorted array one element at a time
d. It divides the list into two halves, sorts each half, and then merges them

The correct answer is: It divides the list into two halves, sorts each half, and then merges them

Question **6**

Complete

Mark 1.00 out of 1.00

What is one advantage of sorting a list before performing a search operation?

a. It makes the search operation slower
b. It allows for faster searching
c. It increases the number of comparisons needed
d. It has no effect on the search operation

The correct answer is: It allows for faster searching

Question **7**

Complete

Mark 1.00 out of 1.00

What is the primary advantage of the divide-and-conquer approach in sorting algorithms?

a. It avoids the need for recursion
b. It allows for efficient parallel processing and sorting of data
c. It simplifies the sorting process by using only one pass
d. It only works on small datasets

The correct answer is: It allows for efficient parallel processing and sorting of data

Question **8**

Complete

Mark 1.00 out of 1.00

What is one of the first steps in a divide-and-conquer algorithm like Merge Sort?

a. Comparing each element with the others
b. Sorting the entire list sequentially
c. Combining sorted sublists
d. Dividing the input into smaller subproblems

The correct answer is: Dividing the input into smaller subproblems

Question **9**

Complete

Mark 1.00 out of 1.00

_____ explain how an algorithm will perform when the input grows larger.

a. Merging
b. Complexity
c. Sorting
d. Searching

The correct answer is:
Complexity

Question **10**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm would be preferred for its divide-and-conquer approach?

a. Linear Search
b. Merge Sort
c. Binary Search
d. Bubble Sort

The correct answer is: Merge Sort

Question **11**

Complete

Mark 1.00 out of 1.00

Which algorithm is efficient for analyzing the frequency distribution of items in a list?

a. Linear Search
b. Quick Sort
c. Merge Sort
d. Bubble Sort

The correct answer is: Merge Sort

Question **12**

Complete

Mark 1.00 out of 1.00

Why is sorting important for selection operations?

a. It slows down the process
b. It makes it easier to select items based on their relationship to the rest of the items
c. It complicates the selection of items
d. It makes the data unsorted

The correct answer is: It makes it easier to select items based on their relationship to the rest of the items

Question **13**

Complete

Mark 1.00 out of 1.00

What is a key disadvantage of Bubble Sort compared to more advanced algorithms like Merge Sort?

a. Bubble Sort is less efficient for large lists
b. Bubble Sort cannot handle duplicate elements
c. Bubble Sort is difficult to implement
d. Bubble Sort does not guarantee sorted order

The correct answer is: Bubble Sort is less efficient for large lists

Question **14**

Complete

Mark 1.00 out of 1.00

The process of placing or rearranging a collection of elements into a particular order is known as

a. Searching
b. Merging
c. Sorting
d. Rearranging

The correct answer is: Sorting

Question **15**

Complete

Mark 1.00 out of 1.00

What is the primary benefit of using sorting algorithms in programming?

a. Makes data harder to manage
b. Provides a basis for other algorithms to work efficiently
c. Makes code execution slower
d. Decreases the efficiency of algorithms

The correct answer is: Provides a basis for other algorithms to work efficiently